Research Assistant: Complete User & Command Guide

A step-by-step handbook for non-programmers

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Welcome

The **Research Assistant** is an autonomous AI assistant for scientific literature. You give it a research idea or a seed paper (PDF), and it automatically:

1. Finds relevant research papers.
2. Reads their bibliographies and extracts citations.
3. Downloads the legal, open-access PDFs of those references.
4. Builds a searchable knowledge base on your machine.
5. Synthesizes related work and inserts LaTeX `\cite{...}` citations into your draft papers.

You do **not** need to know programming or write code to use this tool. You only need to know how to copy, paste, and press **Enter** in your computer's **Terminal**.

> **Note on Purpose & Legal Responsibility:**
> This project is built solely to support researchers and scholars who are not in the business of money-making and are already resource-constrained. The user is solely responsible for any legalities, terms of service, and copyright obligations that may come with retrieving, storing, and citing literature.

Part 1: First-Time Setup (Do this once)

Step 1: Open Terminal and Navigate to the Folder

Open your **Terminal** app (on Linux/Mac, search for "Terminal" and open it).

Type `cd` followed by the path to where you downloaded or extracted this project, then press **Enter**:

```
cd path/to/research_assistant
```

(Tip: You can also type `cd` *with a space and drag-and-drop the project folder from your file manager straight into the Terminal window).*

Step 2: Create a Private Workspace

(Note: The app requires Python 3.10, 3.11, or 3.12. Do not use Python 3.13 because one of the components, lxml, cannot compile on 3.13).

Paste these two lines:

```
python3.12 -m venv .venv
source .venv/bin/activate
```

(Your prompt will now start with `(.venv)`. This confirms you are in your private workspace).

> **CRITICAL RULE FOR TERMINAL USERS:** When you close your Terminal window, it **forgets** that you turned on this private workspace.
>
> **Every single time you open a new Terminal window to use this app, you must activate it again:**
>
> ```
> source .venv/bin/activate
> ```
>
> Always check that `(.venv)` appears at the beginning of your command prompt before running any commands. If you ever see `command not found: streamlit`, it means you forgot this line!

Step 3: Install the Components

Paste this command to install all necessary tools:

```
pip install -r requirements.txt
pip install -e .
```

(This downloads required libraries. It takes 2 to 5 minutes depending on your internet connection).

Step 4: Download the Free AI Models

Make sure Ollama is installed and running on your computer. Then, paste these commands one by one to download the local AI models:

```
ollama pull nomic-embed-text
ollama pull gemma4:e2b-mlx
```

(This is a one-time download of the embedding model and the local language model).

Step 5: Install Docker and Start GROBID

GROBID is the component that reads each paper's reference list properly. Without it the app still runs, but it finds references with no authors, years or DOIs — and it will not warn you that this has happened. Everyone on the project should have it running.

It needs **Docker**. The full walkthrough — installing Docker on your operating system, starting the server, and every error message it can produce — is in the companion handbook:

> **Research Assistant: GROBID Setup Guide** (`Research_Assistant_GROBID_Guide.pdf`)

Once Docker is installed, this is the whole of it:

```
docker pull grobid/grobid:0.8.1
docker run --rm -d --name grobid -p 8070:8070 grobid/grobid:0.8.1
```

Wait about 30 seconds, then confirm it is awake:

```
curl http://localhost:8070/api/isalive
```

(You want to see the word `true`. If you see anything else, check the GROBID guide.)

Part 2: Starting the Web App (Recommended)

The easiest way to use the Research Assistant is via its **visual web browser interface**, where you can click buttons and upload files.

1. Launch the Web Interface

Copy and paste this command block:

```
source .venv/bin/activate
export UNPAYWALL_EMAIL="your.email@example.com"
export CITATION_LLM_MODEL="gemma4:e2b-mlx"
export CITATION_LAYOUT_DETECTION=0
streamlit run app.py
```

> **Note on `CITATION_LAYOUT_DETECTION` (`0` vs `1`):** * **Keep `0` (default for most users):** 95%+ of scientific citations and literature summaries rely entirely on written claims in the text. Setting `0` keeps the pipeline blazing fast, lightweight, and free of heavy GPU/Torch dependencies. * **Use `1` only if:** You specifically need the system to locate, crop, and cite visual charts, plots, or data tables from papers, and you have installed the optional layout stack from `requirements-layout.txt`.

2. Using the Browser App

Your web browser will automatically open at:

`http://localhost:8501`

Inside the browser, you will see four main functional tabs:

- **Tab 1: Research a topic**
  - Type a research idea (e.g., `topological protection in disordered quantum wires`) OR upload your own PDF paper.
  - Click **Build corpus**.
  - Watch the progress live as papers are found, downloaded, and summarized.
- **Tab 2: Cite a draft**
  - Paste a single sentence from your paper (e.g., *Anderson localization suppresses diffusion in 1D.*).
  - Click **Suggest a citation**.
  - The assistant retrieves matching passages and suggests the exact reference with a LaTeX `\cite{...}` tag and an explanation.
- **Tab 3: Cite a whole draft**
  - Upload a `.txt` file containing your full paper draft.
  - Click **Cite the draft**.
  - The assistant inspects every sentence, determines which claims need citations, searches the corpus, and produces:
    1. A cited draft document.
    2. A `_citations.json` file ready for BibTeX.
    3. A sentence-by-sentence evaluation report.
- **Tab 4: Research chat**
  - Have a multi-turn conversation with all the papers in your library.
  - Type questions and receive streaming answers grounded in your downloaded PDFs.

3. Stopping the App

To stop the web app at any time, switch to your Terminal window and press `Ctrl + C`.

Part 3: Command-Line Shortcuts (Typing Commands)

If you prefer to run specific tasks directly from the command line without opening the web browser, use these commands:

1. Research an Idea from Scratch

Searches arXiv/OpenAlex, downloads the seed paper, extracts its citations, downloads available references, and writes a related-work overview:

```
python orchestrate.py --query "topological protection in disordered quantum wires"
↪  --ask
```

2. Research Starting from a PDF File You Have

Seeds directly from a paper you already downloaded:

```
python orchestrate.py --seed-file "/path/to/your/paper.pdf" --ask
```

3. Research Starting from an Online URL

Seeds directly from an arXiv or PDF link:

```
python orchestrate.py --seed-url "https://arxiv.org/abs/2401.12345" --ask
```

4. Find a Citation for a Single Sentence

Suggests a citation for an individual sentence claim:

```
python -m research_assistant.agents.agent4_assistant --text "Anderson localization
↪  suppresses diffusion in 1D."
```

5. Cite an Entire Draft Document

Analyzes every sentence in `my_draft.txt` and writes the cited version to `cited_draft.txt`:

```
python -m research_assistant.agents.agent5_batch_citer --file my_draft.txt --out
↪  cited_draft.txt
```

6. Interactive Chat with Downloaded Papers

Chat with your paper library in the Terminal:

```
python -m research_assistant.agents.agent7_research_chat
```

- Type `/sources` to view the papers that backed the last answer.
- Type `/export` to save the chat transcript to a markdown file.
- Type `exit` to quit.

7. Automatic Background Folder Watcher

Runs quietly in the background. When you drop a PDF into the `data/raw/` folder, it automatically downloads references and adds them to your search library:

```
python watch.py
```

Part 4: Everyday Quickstart (Using the App Again)

Whenever you open a fresh Terminal window in the future, follow this simple 3-step process:

Step 1: Navigate to the folder and turn on `.venv` (MANDATORY)

Because Terminal starts fresh every time, you **must** tell it to activate your private workspace first:

```
cd path/to/research_assistant
source .venv/bin/activate
```

(Always verify that `(.venv)` appears at the start of your line before proceeding).

Step 2: Start GROBID

Docker does not restart GROBID for you after a reboot, so start it again:

```
docker run --rm -d --name grobid -p 8070:8070 grobid/grobid:0.8.1
```

(Already running? `docker ps` will list it, and you can skip this.)

Step 3: Run the App

Now simply start whichever mode you need:

- **To launch the visual web browser app (Recommended):**

  ```
  streamlit run app.py
  ```

- **Or to run a research topic from the command line:**

  ```
  python orchestrate.py --query "your research topic" --ask
  ```

- **Or to chat with your papers in the terminal:**

  ```
  python -m research_assistant.agents.agent7_research_chat
  ```

Part 5: Troubleshooting Guide

What you see	What it means	How to fix it
`command not found: python3.12`	Python 3.12 is not installed.	Install Python 3.12 (e.g. from python.org) or use Python 3.10 / 3.11.

What you see	What it means	How to fix it
Red error mentioning `lxml` during install	You are using Python 3.13.	Delete `.venv` folder and recreate it using `python3.12 -m venv .venv`.
`command not found: streamlit`	The workspace environment is inactive.	Run `source .venv/bin/activate` first.
The app opens but every answer errors	Ollama is not running or model missing.	Open the Ollama app, then run `ollama list` in Terminal to check models.
Red **GROBID** dot in the sidebar	GROBID extraction server is stopped.	Start it: `docker run --rm -d --name grobid -p 8070:8070 grobid/grobid:0.8.1`, or click **Start GROBID** in the sidebar. Do not skip this — the app carries on with much weaker references and does not warn you.
`docker: command not found`	Docker is not installed.	See the **GROBID Setup Guide** — Part 1.
`curl` on port 8070 says `Connection refused`	GROBID is still starting, or is not running.	Wait 30 seconds and retry, then check `docker ps`.
`Address already in use`	The app is already running in another window.	Find that Terminal window and press `Ctrl + C`, or close it.

Hobbes: *Won't inventing a robot be more work than making the bed?*
Calvin: *It's only work if somebody makes you do it.*